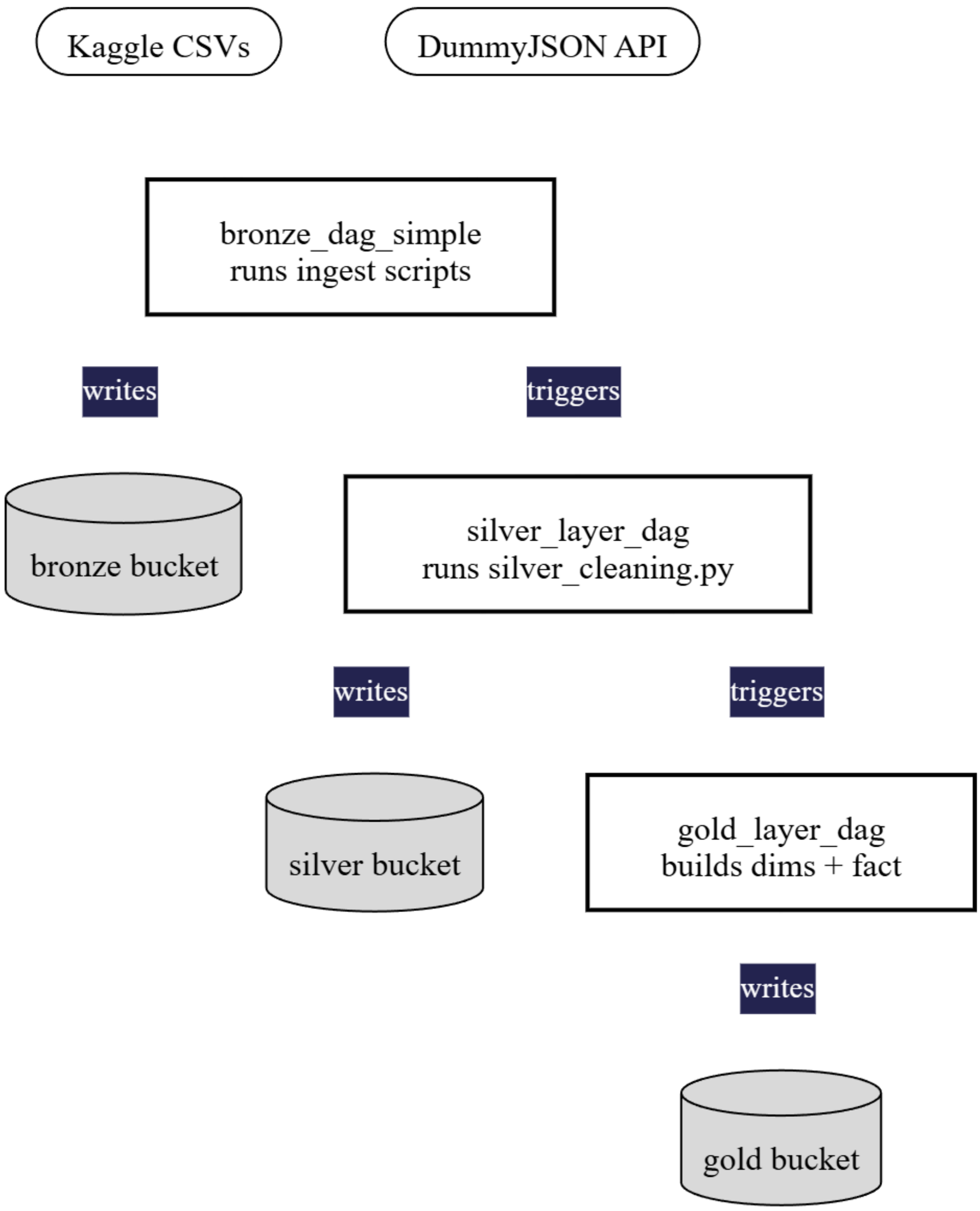
SynthCart E-Commerce Data Engineering Platform

1. Executive Summary

The **SynthCart E-Commerce Data Engineering Platform** is a modern data pipeline implementing the **Medallion Architecture (Bronze → Silver → Gold)** for processing e-commerce data.
Built using **Apache Airflow**, **MinIO**, and **PySpark** in a fully containerized **Docker** environment, the platform automates data ingestion, cleaning, and transformation to deliver analytics-ready datasets.

Project Duration: 3 Weeks
Team Size: 9 Members (3 Data Engineers + 6 Data Analysts)
Status: Completed

2. Architecture Overview



Technology Stack

Component	Technology	Purpose
Data Lake	MinIO	Object storage for all data layers
Orchestration	Apache Airflow	Workflow automation
Processing	PySpark	Data transformation
Database	PostgreSQL	Metadata & data warehouse
Infrastructure	Docker Compose	Container orchestration

Data Flow

Kaggle + APIs → Bronze (Raw) → Silver (Cleaned) → Gold (Analytics-Ready) → Power BI

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3. Team Structure & Responsibilities

Data Engineering Team

Afnan Khan – Bronze Layer & Pipeline Orchestration
• Raw data ingestion from Kaggle and DummyJSON API
• Airflow DAG development and scheduling
• Pipeline automation and monitoring

Farheen Muzaffar & Muhammad Awais – Silver Layer
• Data cleaning and validation
• PySpark transformations
• Data quality checks

Muhammad Awais – Infrastructure & Gold Layer Transformations
• Docker infrastructure setup
• Fact and dimension table creation
• Final data exports

Data Analysis Team

Abdur Rehman, Aqsa Majeed, Ayesha Saleh, Salman Qureshi, Saud Ijaz, Wania Nafees, Zohair Raza

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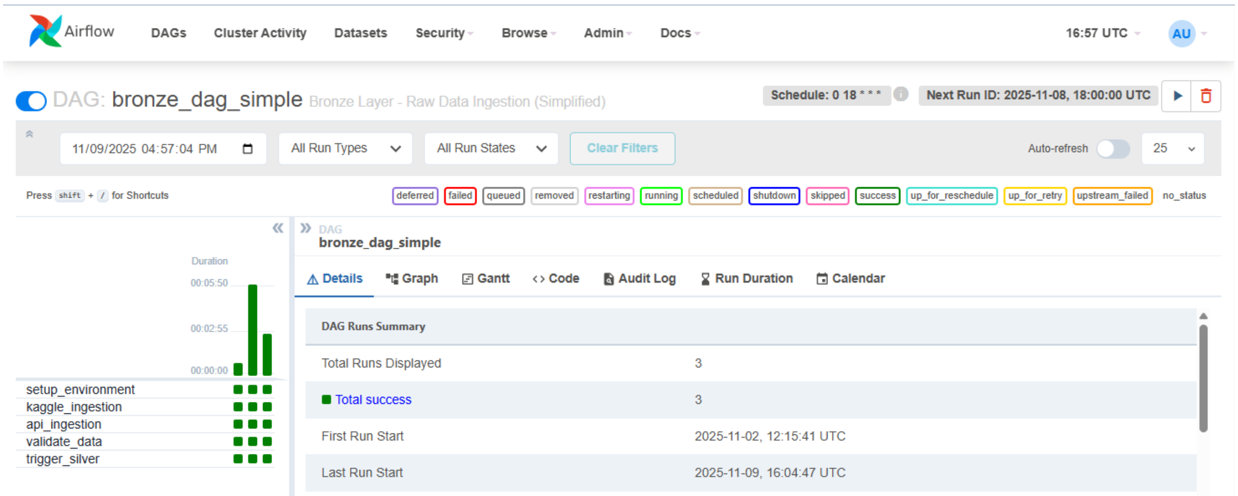
4. Implementation Details

4.1 Bronze Layer – Raw Data Ingestion

Owner: *Afnan Khan*

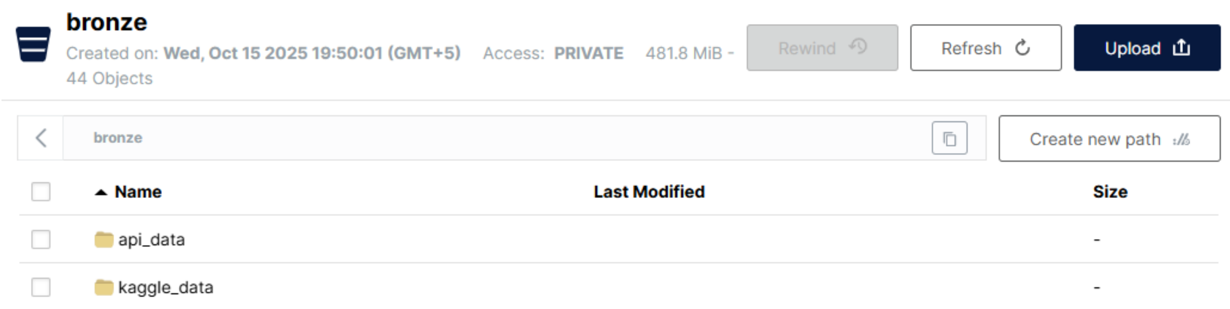


The Bronze Layer ingests raw data from:

- **Kaggle Dataset:** Brazilian E-Commerce (Olist) – 8 CSV files
- **DummyJSON API:** Supplementary product and user data

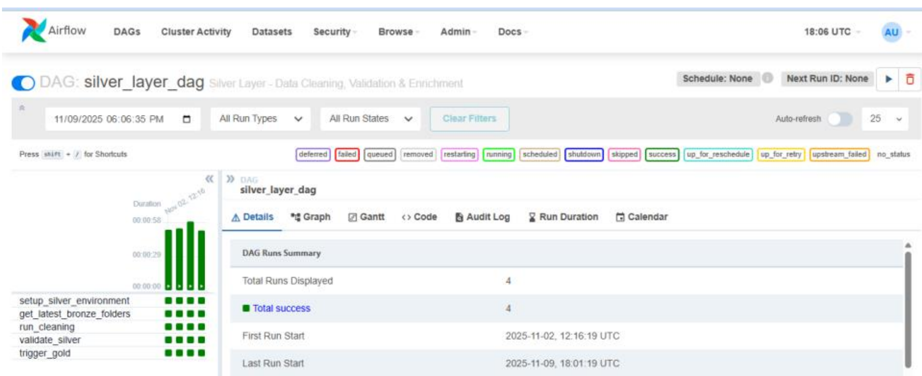
Key Features:

• Automated daily ingestion (6 PM UTC)
• Data stored in MinIO /bronze/ bucket
• Timestamped partitioning for versioning
• Error handling with 2 retries



4.2 Silver Layer – Data Cleaning

Owner: *Farheen Muzaffar & Muhammad Awais*



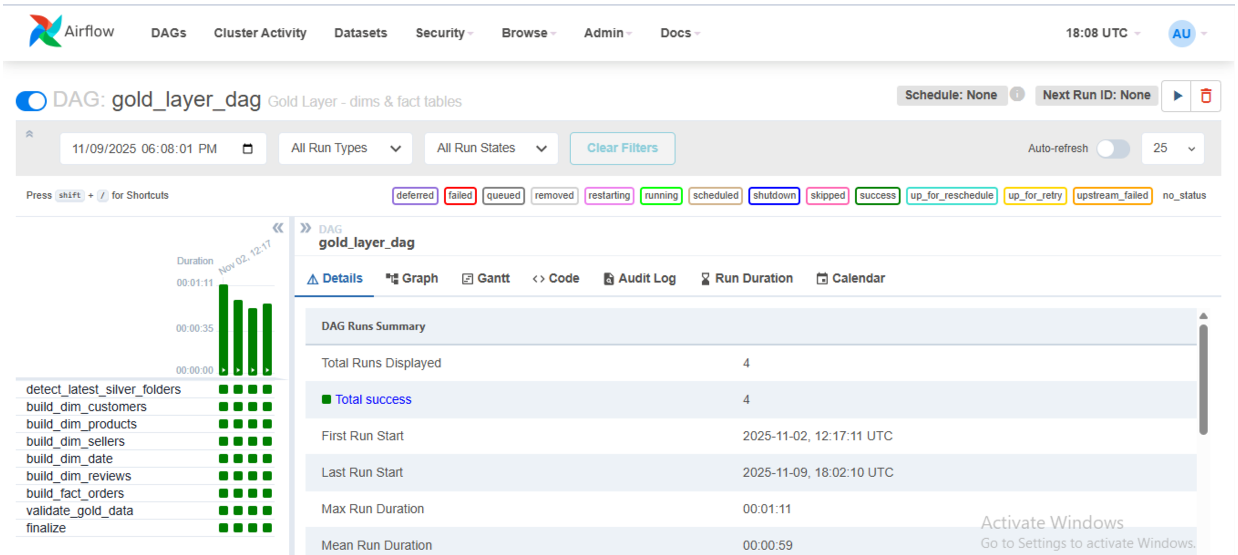
The Silver Layer transforms raw data into cleaned, validated datasets using **PySpark**:

Operations:
• Remove duplicates and handle missing values
• Standardize date formats and text fields
• Validate data types and ranges
• Create derived fields and surrogate keys

Output: Cleaned Parquet files in MinIO /silver/ bucket with **98.5% data quality score**

4.3 Gold Layer – Analytics Tables

Owners: *Muhammad Awais*



The Gold Layer creates a **star schema** with business-ready tables:

Dimensional Model:

- **Fact Table:** fact_orders (transaction-level data)
- **Dimensions:**
  - dim_customers (customer profiles)
  - dim_products (product catalog)
  - dim_sellers (seller information)
  - dim_date (time dimension)
  - dim_reviews (customer reviews)

Deliverables:
• Parquet files in /gold_exports/
• SQL dump (gold_dump.sql)
• Ready for Power BI connection

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5. Deployment & Infrastructure

[SCREENSHOT: Docker Containers Running]

All services run in **Docker containers**:

- Airflow (Webserver, Scheduler, Worker)
- PostgreSQL (Metadata DB)
- Redis (Task Queue)
- MinIO (Object Storage)

Quick Start:

docker-compose up -d

Access Points:

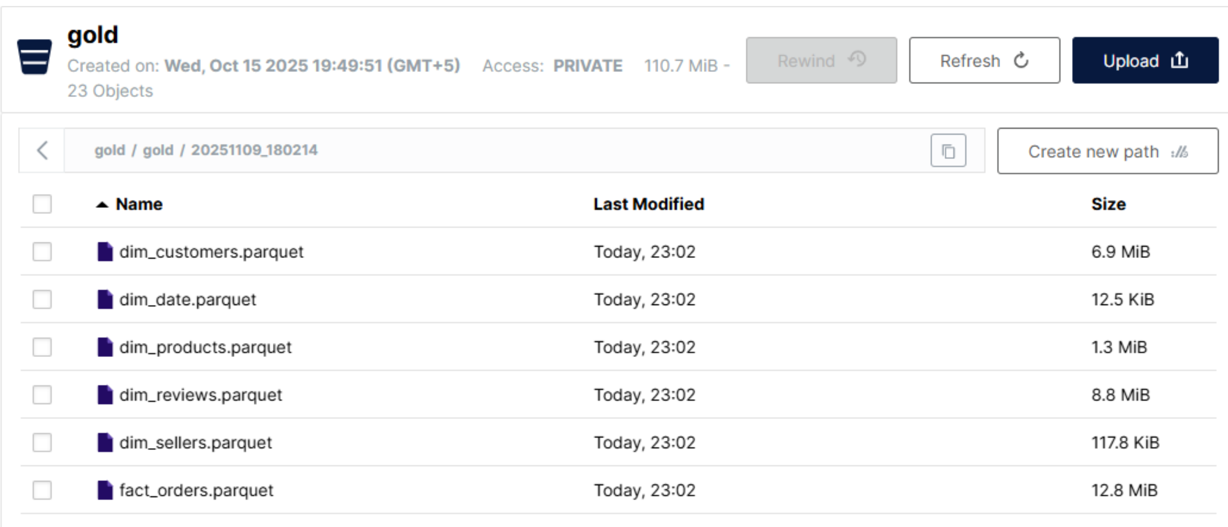
- Airflow UI: http://localhost:8080 (airflow/airflow)
- MinIO Console: http://localhost:9001 (minioadmin/minioadmin)

[SCREENSHOT: Airflow Dashboard]

Final Deliverables

For Data Analysis Team:

1. **PostgreSQL Database** with 6 tables ready for Power BI
2. **Parquet Files** in /gold_exports/ folder
3. **SQL Dump** for portable backup
4. **Documentation** with connection details



7. Challenges & Solutions

Challenge	Solution	Impact
PySpark needs Java	Custom Dockerfile with OpenJDK 17	Enabled distributed processing
Large file handling	Streaming uploads & chunked processing	70% memory reduction
Manual DAG triggering	TriggerDagRunOperator for automation	Fully automated pipeline
Data quality issues	Comprehensive validation in Silver Layer	98.5% quality score

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8. Conclusion

The **SynthCart platform** successfully demonstrates a **production-grade data pipeline** using modern tools and best practices.

✅ Fully automated ETL pipeline
✅ High data quality (98.5%)
✅ Scalable containerized infrastructure
✅ Analytics-ready datasets for DA team
✅ Complete documentation and handoff

Business Impact: Saves 20+ hours per week in manual data preparation and enables self-service analytics.

Future Enhancements

- Incremental loading for faster processing
- Real-time streaming with Kafka
- Machine learning integration
- Cloud migration (AWS/Azure)

Contact & Repository

GitHub: https://github.com/awaisstack/synthcart-ecommerce-platform

Data Engineering Team:

- Afnan Khan (Bronze Layer & Orchestration)
- Muhammad Awais (Infrastructure & Gold Layer)
- Farheen Muzaffar & Muhammad Awais (Silver Layer)